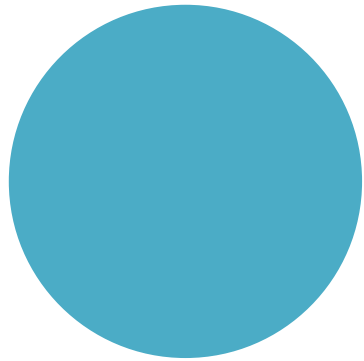


QXU4006

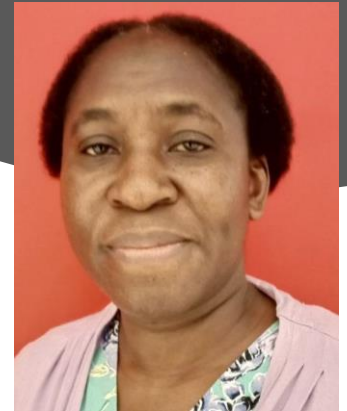
Materials Science 2

Processing and Applications

Classroom Assistants:

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The Sinking of the Titanic



Space Shuttle Challenger Disaster



**Tesla's Shattered
Cybertruck Windows**

Aims of module

Relation between **Structure and Properties**

To provide *an overview* of properties, applications and processing procedures of materials

- Metals
- Polymers
- Ceramics

Select materials for specific applications and processing methods by using the following *material properties*

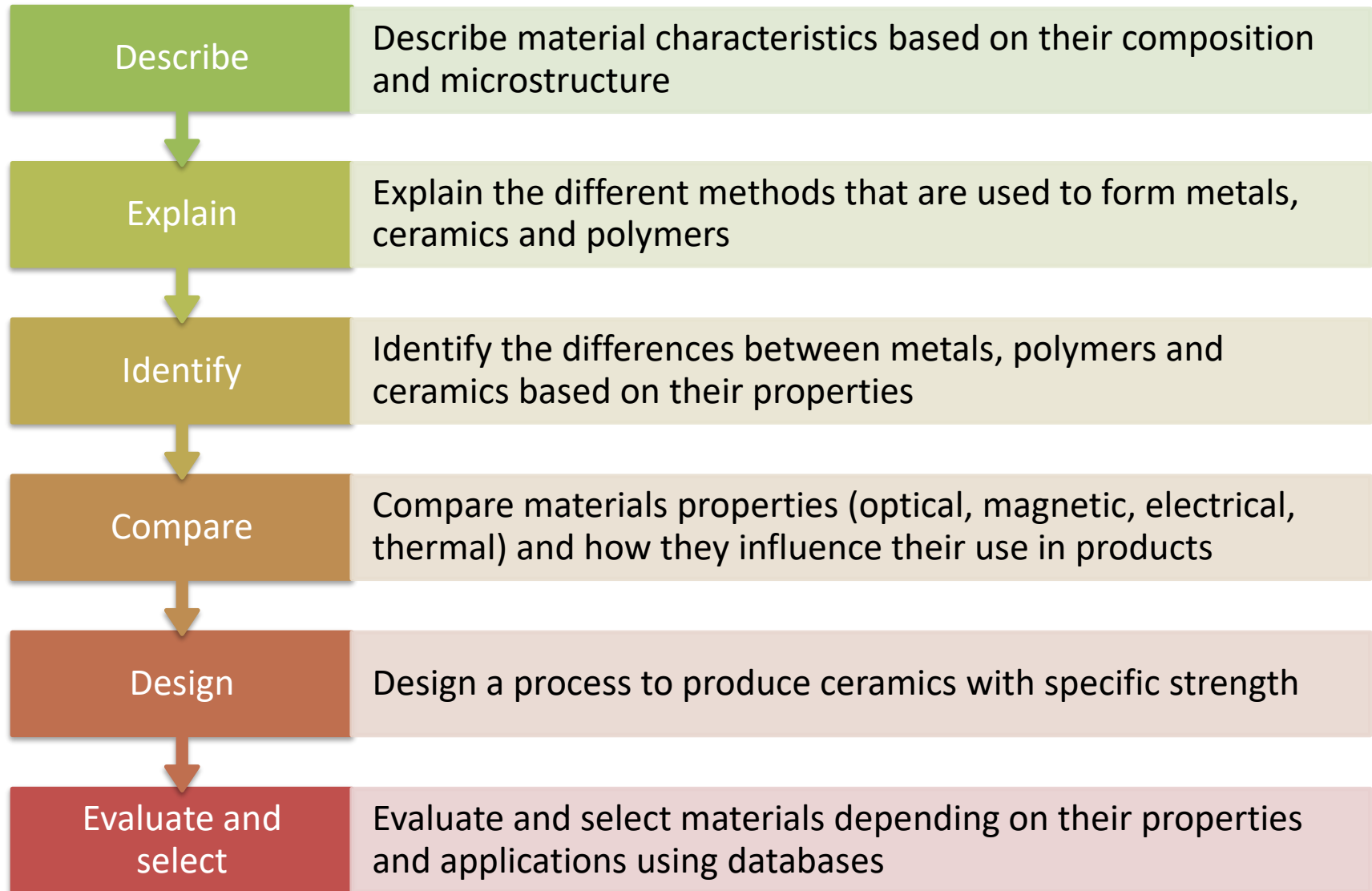
- Electrical
- Thermal
- Magnetic
- Optical

Select materials for applications

At the end of this module you will be able to...



Learning Outcomes

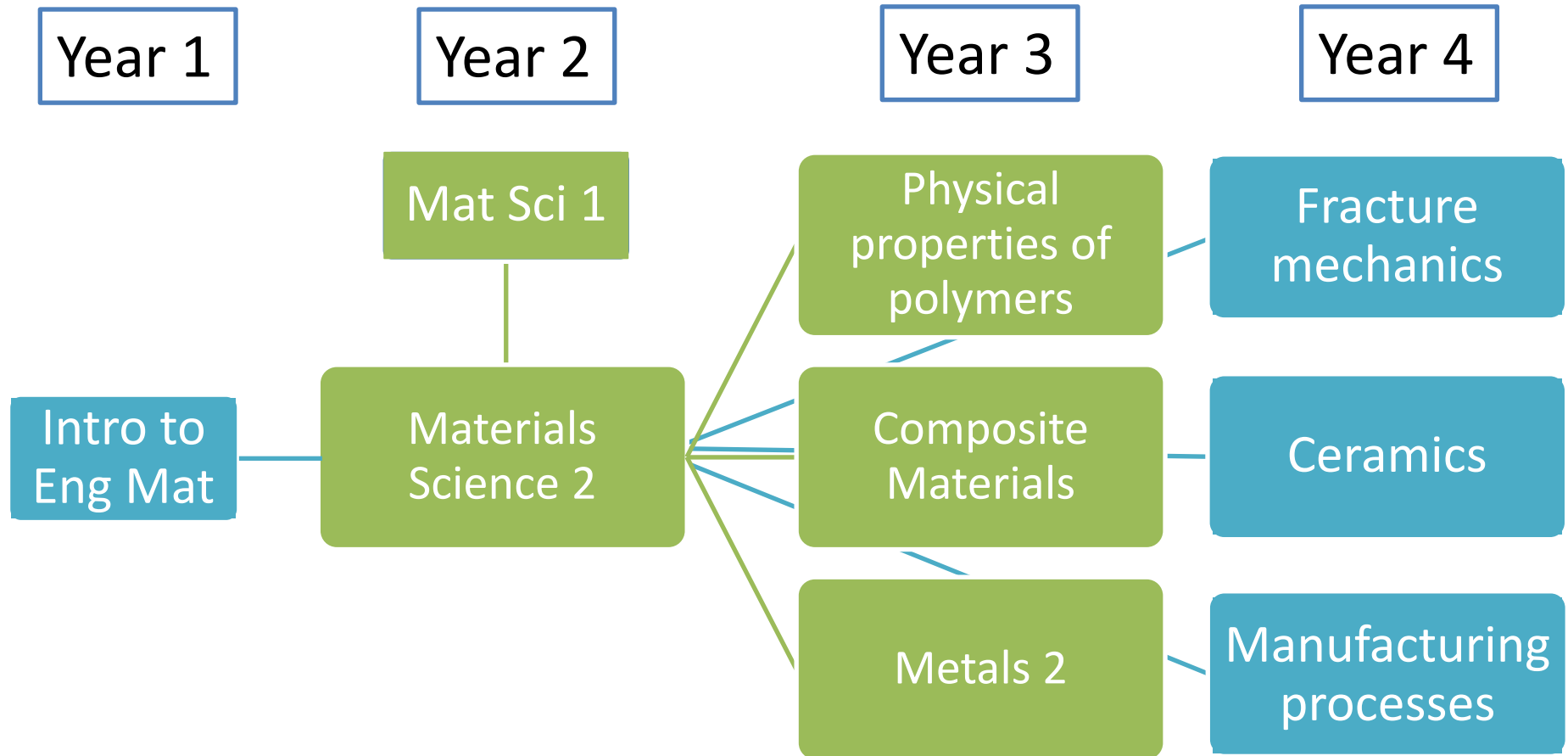


Materials Engineering Programme

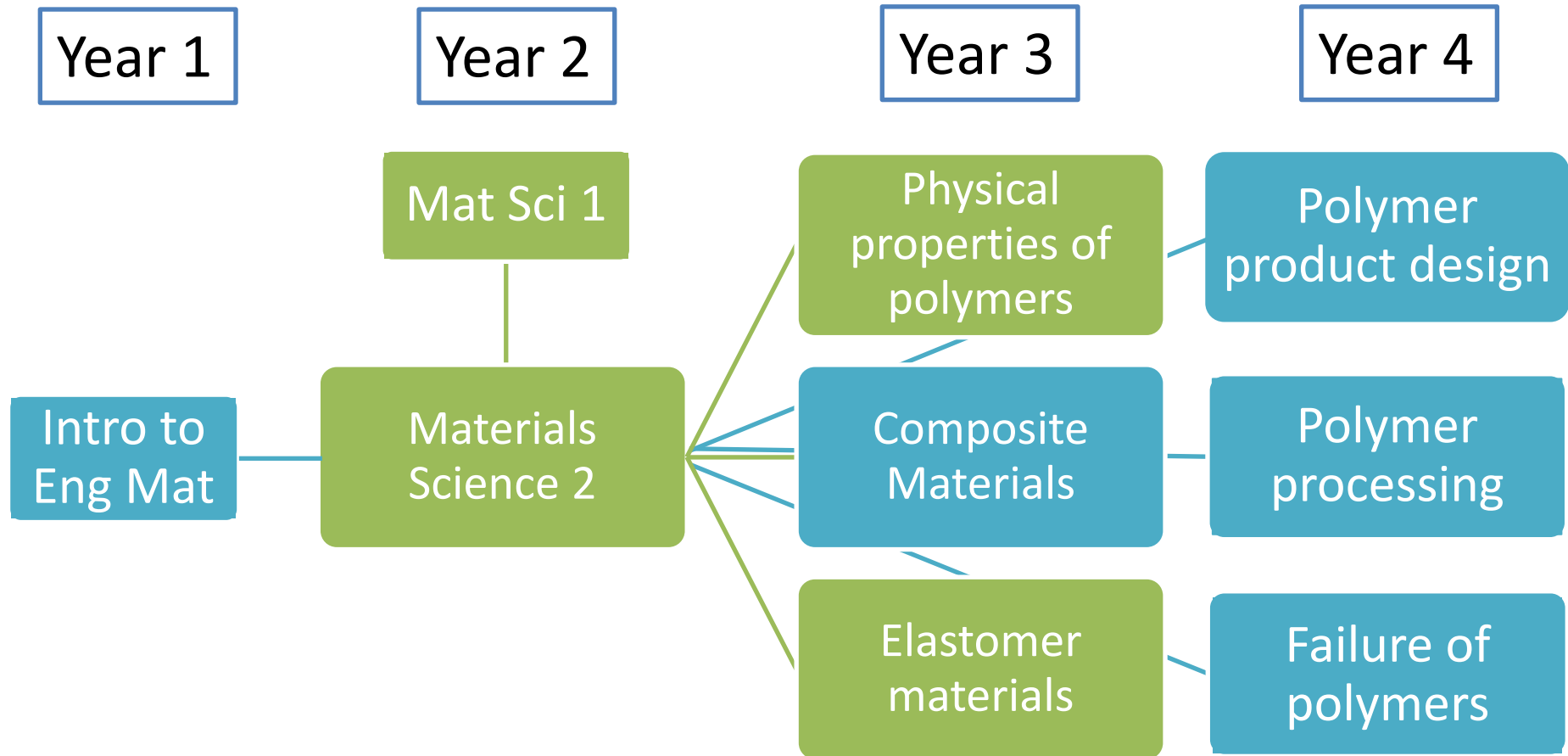


Year 1	Year 2	Year 3	Year 4
Advance Maths 1	Materials Sci 1	Properties of Polymers	Materials Selection
Linear Algebra	Eng Design Methods	Surfaces & Interfaces	Fracture mechanics
General Physics	Experiments in Materials 1	Intro to Functional Materials	Ceramics
Intro to Engineering Materials	Chemistry for Materials	Metals 1	Manufacturing processes
English 1	Molecules to Materials	Composite Materials	
Advanced Maths 2	Materials Sci 2	Experiments in Materials 2	Environmental properties
Maths, Modelling & computing	Mechanical modelling	Structural Characterisation	Fatigue & creep failure
English 2	Thermo & Phase transformations	Metals 2	Renewable energy Materials
PDP1	PDP2	PDP3	Materials Project

Knowledge Progression



Knowledge Progression





Why do I need to study this module?

**WHAT YOU LEARN IN THIS MODULE
CONTRIBUTE TO YOUR STUDENT AND
PROFESSIONAL FUTURE**

Skills gained

Engineering skills

- Knowledge of metals, ceramics and polymers properties
- Select metals, ceramics and/or polymer materials for applications based on their properties
- Design and application
- Develop material processing methods
- Synthesis and evaluation

Practical skills

- Time management
- Organisation
- Team-work
- Communication using technical terminology
- Problem solving
- Ability to identify, integrate and use different resources (Research)
- Planning

Beyond University ... jobs



Understanding
Materials
Properties
Processing
Applications

The role

Joining RINA as a Chemical Regulatory Product Specialist, you'll help our customers place products on the market, which meet all required technical and legal requirements (including the REACH regulation, RoHS directive, POPS and similar global measures) related to design, manufacture and supply of engineering products - primarily from the perspective of substance requirements in hardware. The Chemical Regulatory Product Specialist role provides the opportunity for you to build your skills and relevant experience to date

Beyond University ... jobs

Research Scientist – Optical Physics Of Materials

Surrey NanoSystems - Newhaven BN9

Apply Now



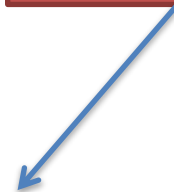
As a graduate research scientist, you will be joining our team of physicists and chemists who work across a variety of programs involving the creation of super-black absorber coatings. Specifically, you will be involved with optimisation of optical properties of Vantablack based materials from UV to Microwave spectrum.

This is a hands-on role with a great potential for new scientific learning and creativity. It is the perfect opportunity to make a difference in a role where the training is done on the job, where every week brings a new and exciting challenge.

Key requirements to apply:

- Academic background in optical properties of materials, optics, physics, condensed matter physics, Materials science.

Knowledge of
Materials
Optical
Properties



How we will study

Welcome to QXU4006 !

- Delivery of the module will be through :
 - 'Live' Lectures to your classroom
 - In-class discussions & exercises
 - In-class quizzes
 - Group study on application of materials

How we will work

- I will join you in the class (live session)
- Quiz in class
 - QMES2019 Mat
 - QMES2019 Poly
- Ask questions **in class** using WeCom
- Ask questions **after class** on QMPlus Module Forum

Where to find information

QMPlus

- Module information and announcements
- Lecture notes
- Coursework instructions
- Coursework submissions

- **Module Forum** – ask and answer questions (after class)
 - Everyone can see your question
 - Everyone can see the answer

Where to find background reading

We will follow the chapters in the book:

W.D. Calister and D.G. Rethwisch
9th Edition, SI version

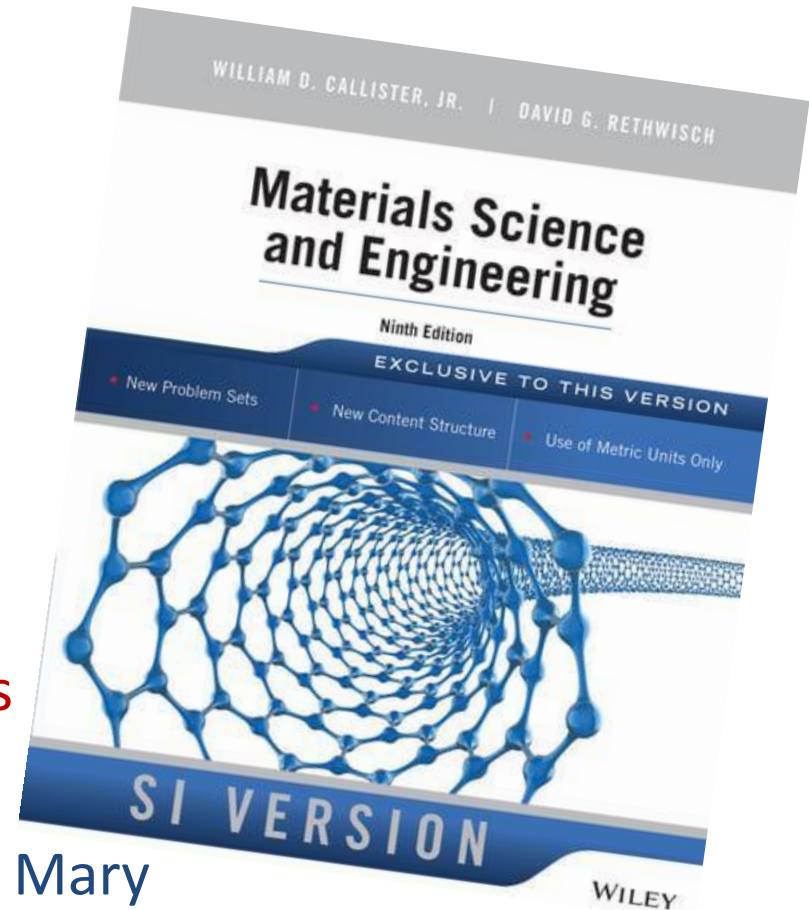
Chapters 13-15, 17

Chapters 19-23

Read the chapter before the class

E-book available from the Queen Mary

Library and hard copies available from the NPU office





Module content

Content

- Properties and Applications of metals combined with Fabrication and processing of metals
- Characteristics and applications of polymers combined with Fabrication and processing of polymers
- Properties and application of Ceramics combined with Fabrication and processing of ceramics

Chapter No in Callister book (9th SI version)

- 13 and relevant parts for metals from 17
- 15 and relevant parts for polymers from 17
- 14 and relevant parts for ceramics from 17

NOTE: You can use ANY version of the Callister book – the content is the same, but the Chapter number is different, just check that the title of the chapter is comparable to the content here.



Module content

Content	Chapter No in Callister book (9 th SI version)
■ Electrical properties	■ 19
■ Thermal properties	■ 20
■ Magnetic properties	■ 21
■ Optical properties	■ 22
■ Economic, Environmental and societal issues in Materials Science	■ 23

NOTE: You can use ANY version of the Callister book – the content is the same, but the Chapter number is different, just check that the title of the chapter is comparable to the content here.

Course organisation



Before class

Reading material

Videos

Material in QMPlus



During class

Lectures

Group work

Individual work –
calculations



After class

Review of material

Q&A – Module forum

Attendance and Engagement



Attendance will be monitored by

Joining Zoom/Tencent meeting for every lesson

Completion of activities in QMPlus (i.e. quiz)

Joining live chats when required



Engagement will be monitored using

User track activity in QMPlus

Downloads of documents from QMPlus

Videos watched in QMPlus

Submission of worksheet or coursework in QMPlus

Module assessment



Worksheets: 3 in-class exercises (10% each)



Group work (30%)



Online tests (20%)



Exam (20%)

- Continuous and Summative assessments
- It is designed to get you engaged continuously with the content material
- It requires independent study and collaboration with other students
- For specific deadlines refer to QMplus



Remember

Completing all the assessed and not-assessed examples and reviewing material as suggested is the only way to achieve a good mark in this module

Requirements to be successful in this module



Individual work

Independent review of content

Prepare for class

Research concepts, technical and English words

Explore content in other books and electronic resources



Group work

Reach out to other students using electronic media

Organise tasks and time. Be clear of who is doing what

Meet and communicate frequently

Keep a record of discussions

BE
CURIOUS

WORK
HARD

BE
PATIENCE

Thank you!



There is an ancient Chinese proverb:

*“Teachers open the door
You enter by yourself”*

